

Engineering a Digital Twin for the Monitoring and Control of Beer Fermentation Sampling

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Context

During beer brewing,

beer fermentation is a crucial step

Complex chemical process over a few weeks:



Beer Monitoring during Fermentation

Measurements:

- temperature
- dissolved oxygen
- pH
- conductivity
- specific gravity (alcohol content)

Usually manually performed

Issues:

- **Slow** (45 minutes per batch per day)
- **Contamination**
- **Wastes product** by sampling

Addressed by:

1) Building **physical hardware**, and
2) Engineering **DT for monitoring/control**
   Improves quality of fermentation

Beer Sampler



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UT VIS SIC FERMENTUM

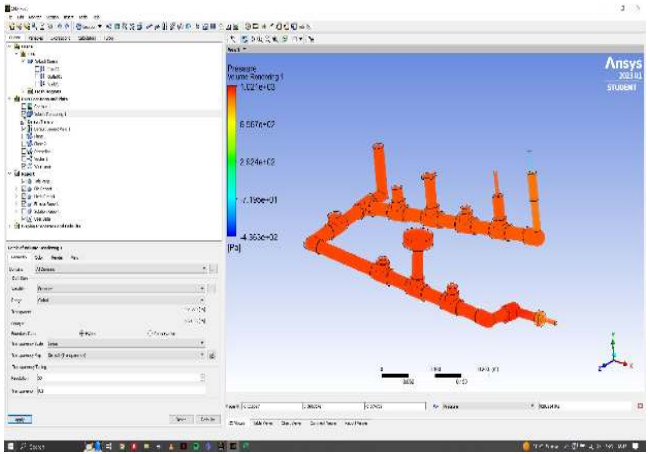
Beer-brewing club at Polytechnique Montréal

Sampler has:

- Pressurized measurement chamber with Arduino-based control system
- Connects to fermentation tank to:
  - Automatically extract liquid
  - Perform measurements
  - Return liquid to tank

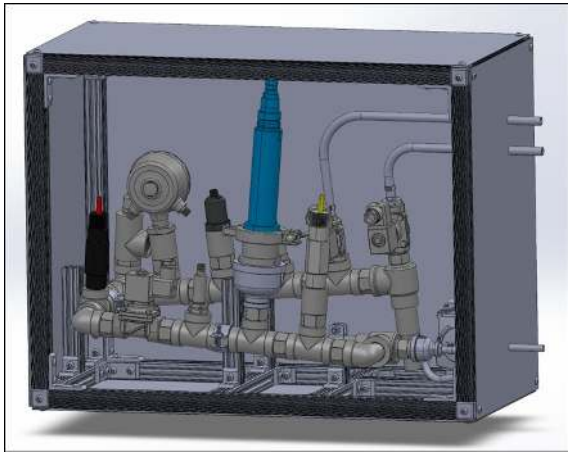
Engineering Timeline

Fall 2023 - Simulation



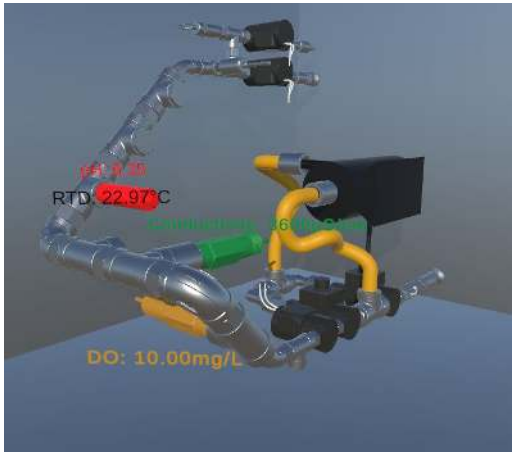
Tools:
Ansys Flow

Fall 2024 - Geometry



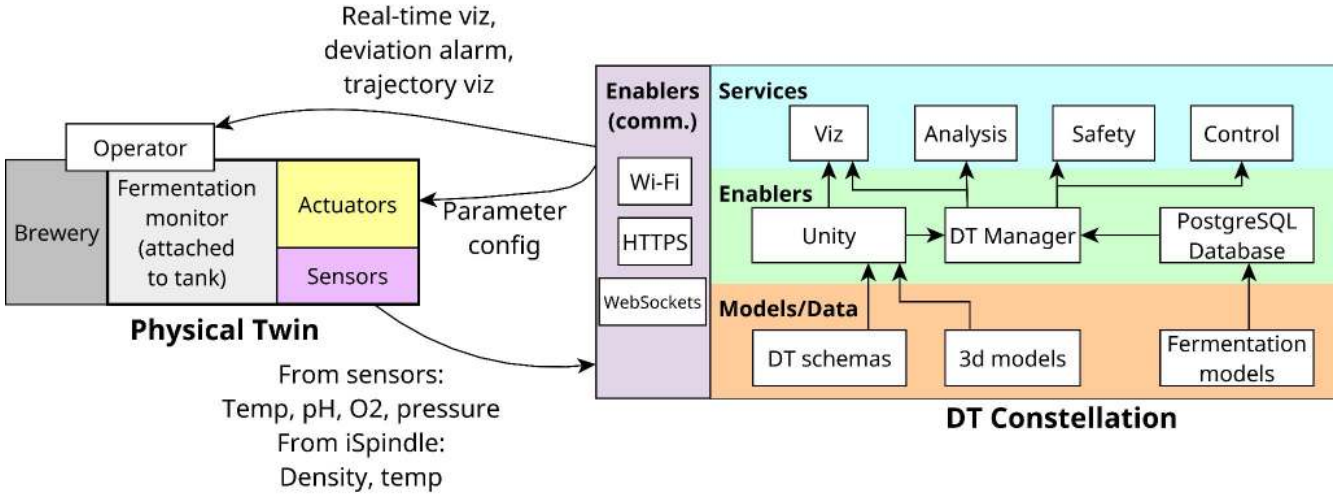
Tools:
CAD

Fall 2025 - DT



Tools:
Unity

DT Constellation



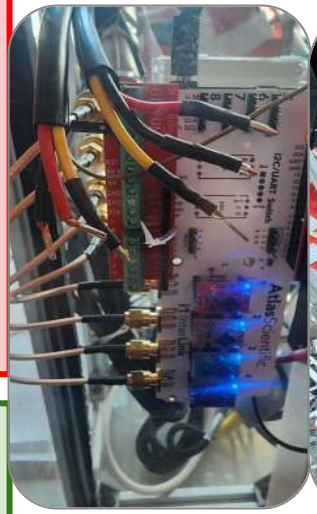
Services:

- **Viz** is real-time/historical dashboards and 3D (Unity)
- **Analysis** is trend analysis, calculation of derived metrics
- **Safety Monitoring** is real-time pressure monitoring
- **Control** is parameter & data collection frequency setting on sampler

Gil et al (2024). Toward a systematic reporting framework for digital twins, *Simulation*.

Physical Challenges & Solutions

- **Dangerous** system
  - Seven bar = three times tire pressure
- **Wet** system
  - Sensor failure in these connections
  - Leakage under pressure
- **Noisy** system
  - Sensor noise
  - Electrical noise





- **Iterative development** at low pressures
- Hardware and software **pressure valves**
- Add **electrical shielding and filtering** in hardware and software

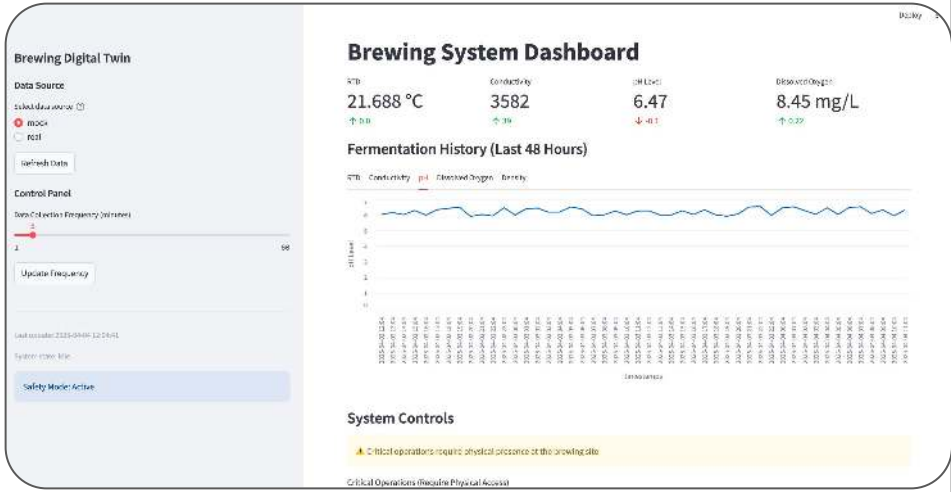
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Software Challenges & Solutions

IoT challenges:

- Network **latency**, clock **drift**
- Viz and database **performance**

- **Latency, buffering** algs
- Improve performance with **selective updates** and **optimizations**

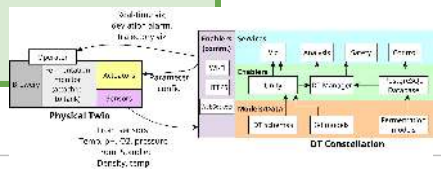


Interdisciplinary Challenges & Solutions

Project Domains:

- Brewing science
- Electrical engineering
- Software development
- Mechanical engineering
- Process control

- Regular meetings
- Common vocabulary
- DT constellation





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Results

- **Manual sampling time reduced** from 45 min per batch per day to 4 min (91% reduction)
- 23 **automated alerts**
- Five-second collection intervals = **17,280 data points per day** vs 8 manual readings before
- Over **500 pressure cycles** without incident

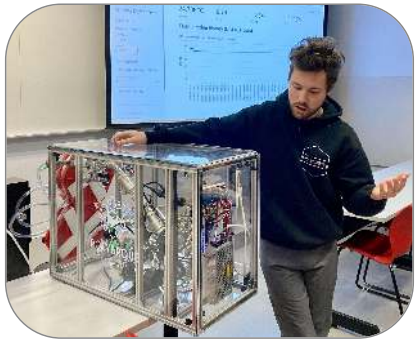
Lessons Learned

1) **Safety-first is essential**
   a) Progressive implementation of hardware and software
   b) Safety valves and shutoffs in hardware and software/DT

2) **Common constellation framework/visual language** significantly reduced cross-domain misunderstandings

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Conclusion



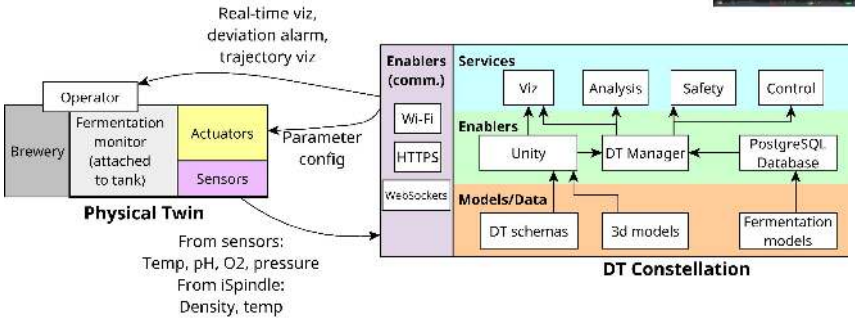
Beer fermentation sampler, automatically sampling and sensing various parameters

Beer Fermentation DT

Paper - Engineering a Digital Twin for the Monitoring and Control of Beer Fermentation Sampling
Presentation - Report



'True' DT with **viz**, **analysis**, **safety** and **control services**



Engineering a Digital Twin for the Monitoring and Control of Beer Fermentation Sampling

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https://bentleyjoakes.github.io/dte_course/